UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Class 004 · Particular integral Type 2 (sin ax, cos ax) · compiled 16 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S004·9	Unit 2 deck 004 (“Particular integral of Type 2”), slide 9 (rendered .pptx)
T004 00:10:32	Unit 2 AV-summary transcript “Particular integral of standard functions – Type 2” (24.7 min), segment start
T004b 00:08:13	Unit 2 AV-summary transcript “Particular integral Type 1 and Type 2” (12.6 min, newer recording); its Type 1 part is tagged T004a in the class 003 pack
CW U2 P13	Unit 2 class-work problem 13
ESA Dec-23 3a	Past paper: UE22MA141A Dec 2023, question 3.a
ESA Jan-24 s1 2d	Past paper: UE23MA141A Jan 2024 set 1, question 2.d

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A Formula sheet

1 The Type 2 rule: replace D^2 by $-a^2$

E1.1 · Type 2 rule

ESA ★★★

$$\frac{1}{f(D^2)}\sin(ax+b) = \frac{\sin(ax+b)}{f(-a^2)}$$
$$\frac{1}{f(D^2)}\cos(ax+b) = \frac{\cos(ax+b)}{f(-a^2)}$$

GIVES The PI of a sine or cosine.

APPLIES $f(-a^2) \neq 0$, i.e. $\pm ai$ not roots of the AE.

NOTE Replace D^2 by $-a^2$, not by $(-a)^2$; b plays no part.

S004·4 · S004·15 · T004 00:00:58 · T004b 00:06:32

E1.2 · Writing odd powers through D^2

ESA ★★★

$$D^3 = D^2 \cdot D \to -a^2 D, \qquad D^4 = (D^2)^2 \to a^4, \qquad D^5 = (D^2)^2 D \to a^4 D$$

GIVES A denominator of the form $p + qD$.

APPLIES $f(D)$ has odd powers of D.

NOTE Lecture example: $D + D^2 + D^3 \to D - a^2 - a^2 D$.

T004 00:01:42 · T004 00:11:10 · S004·9

E1.3 · Rationalising a linear denominator

ESA ★★★

$$\frac{1}{p+qD}X = \frac{(p-qD)}{p^2-q^2D^2}X = \frac{(p-qD)X}{p^2+q^2a^2}$$

GIVES A PI with no D left in the denominator.

APPLIES After E1.2 a single D remains; $X = \sin(ax+b)$ or $\cos(ax+b)$.

NOTE $p - qD$ then acts on X: $D\sin(ax+b) = a\cos(ax+b)$, $D\cos(ax+b) = -a\sin(ax+b)$.

S004·9 · S004·17 · T004 00:12:01 · T004b 00:11:14

2 Case of failure: $f(-a^2) = 0$

E2.1 · First case of failure

ESA ★★★

$$f(-a^2) = 0: \quad \frac{1}{f(D^2)} \sin(ax+b) = x\,\frac{\sin(ax+b)}{f'(-a^2)} \quad (\text{same for } \cos)$$

GIVES The PI when the frequency is in the CF.

APPLIES $f(-a^2) = 0$, $f'(-a^2) \neq 0$.

NOTE f' = derivative w.r.t. D; it usually leaves a D (e.g. $2D$), which is then integration $\frac{1}{D}$.

S004·5 · S004·15 · S004·16 · T004 00:03:07 · T004b 00:07:32

E2.2 · Standard results for $D^2 + a^2$

ESA ★★★

$$\frac{1}{D^2+a^2}\sin ax = -\frac{x\cos ax}{2a}, \qquad \frac{1}{D^2+a^2}\cos ax = \frac{x\sin ax}{2a}$$

GIVES The failure PI in one line.

APPLIES $f(D) = D^2 + a^2$ (or this factor, with the rest applied first).

NOTE From E2.1: $\frac{x}{2D}\sin ax = \frac{x}{2}\int \sin ax\,dx$.

ADDED FOR CLARITY S004·16 · T004 00:23:17

E2.3 · Second case of failure

ESA ★

$$f'(-a^2) = 0: \quad \frac{1}{f(D^2)} \sin(ax+b) = x^2\,\frac{\sin(ax+b)}{f''(-a^2)}$$

GIVES The PI when $D^2 + a^2$ is a repeated factor.

APPLIES $f(-a^2) = f'(-a^2) = 0$, $f''(-a^2) \neq 0$.

NOTE And so on: x^3 and f''' if f'' also fails.

S004·6 · S004·15 · T004 00:04:23 · T004 00:05:06

3 Products, powers and sums: reducing to Type 2

E3.1 · Product to sum

ESA ★

$$\begin{aligned}\sin A\cos B &= \tfrac{1}{2}[\sin(A+B)+\sin(A-B)]\\ \cos A\cos B &= \tfrac{1}{2}[\cos(A+B)+\cos(A-B)]\\ \sin A\sin B &= \tfrac{1}{2}[\cos(A-B)-\cos(A+B)]\end{aligned}$$

GIVES Single sines and cosines.

APPLIES $X = \sin ax\cos bx$, $\cos ax\cos bx$, $\sin ax\sin bx$.

NOTE Only the first identity is spoken in the lecture; the other two are standard.

S004·7 · T004 00:05:42

E3.2 · Squares

ESA ★★★

$$\sin^2 ax = \frac{1 - \cos 2ax}{2}, \qquad \cos^2 ax = \frac{1 + \cos 2ax}{2}$$

GIVES A constant (Type 1 with $a = 0$) plus a Type 2 term at frequency $2a$.

APPLIES $X = \sin^2 ax$ or $\cos^2 ax$.

NOTE $\cos^2 \frac{x}{2} = \frac{1+\cos x}{2}$, $\sin^2 \frac{x}{2} = \frac{1-\cos x}{2}$.

S004·7 · S004·13 · S004·17 · T004 00:06:23 · T004 00:20:36 · ESA Jan-24 s3 2c · ESA Jan-24 s4 2c

E3.3 · Cubes

ESA ★

$$\sin^3 A = \frac{3\sin A - \sin 3A}{4}, \qquad \cos^3 A = \frac{3\cos A + \cos 3A}{4}$$

GIVES Two Type 2 terms, at frequencies a and $3a$.

APPLIES $X = \sin^3 ax$ or $\cos^3 ax$.

NOTE Listed on slide 004·7 without the identity.

ADDED FOR CLARITY S004·7 · T004 00:06:41

E3.4 · Superposition

ESA ★★★

$$\frac{1}{f(D)}(X_1 + X_2 + \dots) = \frac{1}{f(D)}X_1 + \frac{1}{f(D)}X_2 + \dots = \mathrm{PI}_1 + \mathrm{PI}_2 + \dots$$

GIVES The PI of a sum of terms of different types.

APPLIES X is a sum, e.g. $e^x + \cos^2 \frac{x}{2}$.

NOTE Each term chooses its own rule (Type 1, Type 2, failure).

S004·11 · S004·14 · T004 00:17:10 · ESA Dec-23 3a

E3.5 · Factor by factor

ESA ★★

$$\frac{1}{\varphi(D)\psi(D)}X = \frac{1}{\psi(D)}\left[\frac{1}{\varphi(D)}X\right]$$

GIVES A way round a failure in one factor.

APPLIES Apply the non-failing factor first, then the other.

NOTE Slide 004·13: $\frac{1}{(D-1)^2}$ on $\cos x$ first, then $\frac{1}{D^2+1}$.

S004·12 · S004·13 · T004 00:18:17 · T004 00:21:50

B Symbol glossary

Symbol	Name	Meaning	Appears in
D	Differential operator	$\frac{d}{dx}$; $\frac{1}{D}$ = integration without a constant	E1.2, E2.1
$f(D)$, $f(D^2)$	Operator polynomial	$f(D^2)$: $f(D)$ written through D^2 where possible	E1.1
$f(-a^2)$	Replaced denominator	Every D^2 replaced by $-a^2$ (minus a squared)	E1.1, E2.1
$f'(D)$, $f''(D)$	Derivatives w.r.t. D	Used in the failure cases; slide 004·15 writes $f'(a)$	E2.1, E2.3
a	Frequency	In $\sin(ax+b)$; not the exponent of Type 1 (same letter)	E1.1
b	Phase constant	Never enters the rule	E1.1
$p+qD$	Linear remainder	What is left after $D^2 \to -a^2$; rationalised with $p-qD$	E1.3
PI_1, PI_2	Partial PIs	PIs of the separate terms of X	E3.4
X	Right-hand side	A function of x; capital X, small x	E1.1

Remark. On the slides $f(D^2)$ means $f(D)$ written so that D appears through D^2 wherever possible; $f(-a^2)$ means "replace every D^2 by $-a^2$". $-a^2$ is minus (a squared), not $(-a)^2$: for $a=2$ it is -4. In the failure rule, f' is the derivative of $f(D)$ with respect to D (so D^2+3 gives $2D$), and the result is still an operator to apply, not a number (slide 004·16). The letter a is the frequency in $\sin(ax+b)$; b is a phase and never enters the rule.